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[GS PAPER III: SCIENCE AND TECHNOLOGY; INTERNAL SECURITY. • GS PAPER II: GOVERNANCE.]
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Satellite Broadband: Markets, Security and Space Sustainability



Explore how satellite broadband is reshaping digital markets while raising challenges of national security and regulatory oversight. Understand the need to balance affordable connectivity and fair competition with responsible satellite deployment and space sustainability.

Satellite Broadband: Markets, Security and Space Sustainability

Missed Part 1?

Read here - Satellite Broadband India's digital Frontier

Syllabus areas:

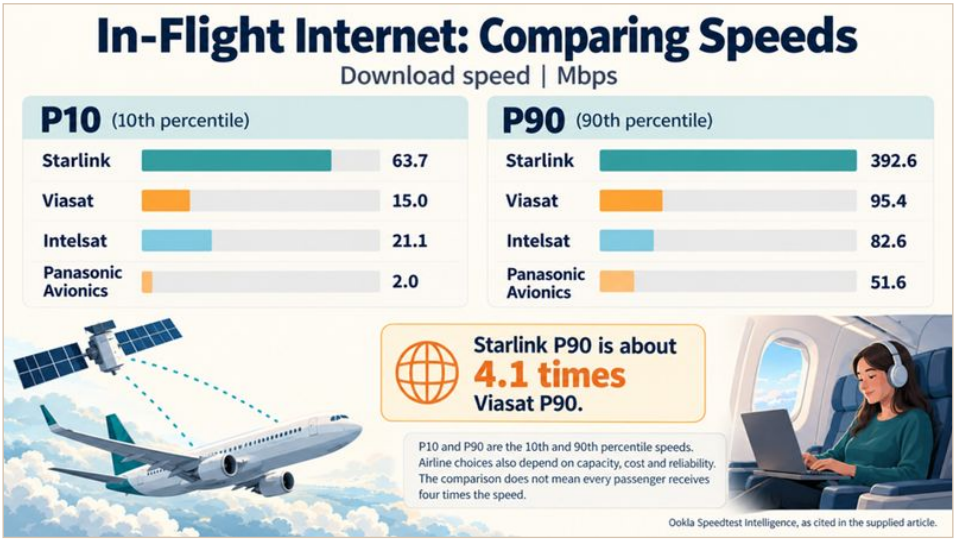
GS Paper III: Science and Technology; Internal Security.

GS Paper II: Governance.

Aviation as an Application of Satellite Internet

Satellite communication enables internet access aboard aircraft. On board equipment connects with satellites, while Wi-Fi distributes connectivity to passengers.

Airlines evaluate providers on **speed, available capacity, reliability, installation requirements and cost**.

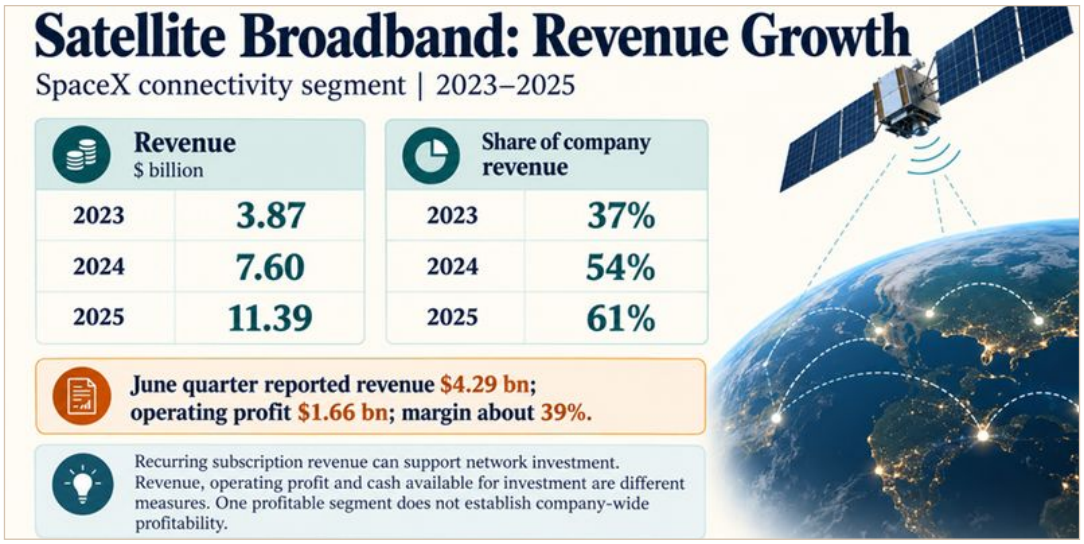


Mbps means megabits per second. **P10 and P90** are the 10th and 90th percentile measurements: approximately 10% and 90% of recorded speeds, respectively, fall at or below these values.

Starlink's P90 figure is approximately four times Viasat's. This does not mean every passenger receives four times the speed. Performance depends on passenger demand, available capacity and operating conditions.

Economics of Commercial Space Services

Satellite broadband generates **recurring subscription revenue**, which can support investment in satellite replacement, network expansion and technology development.



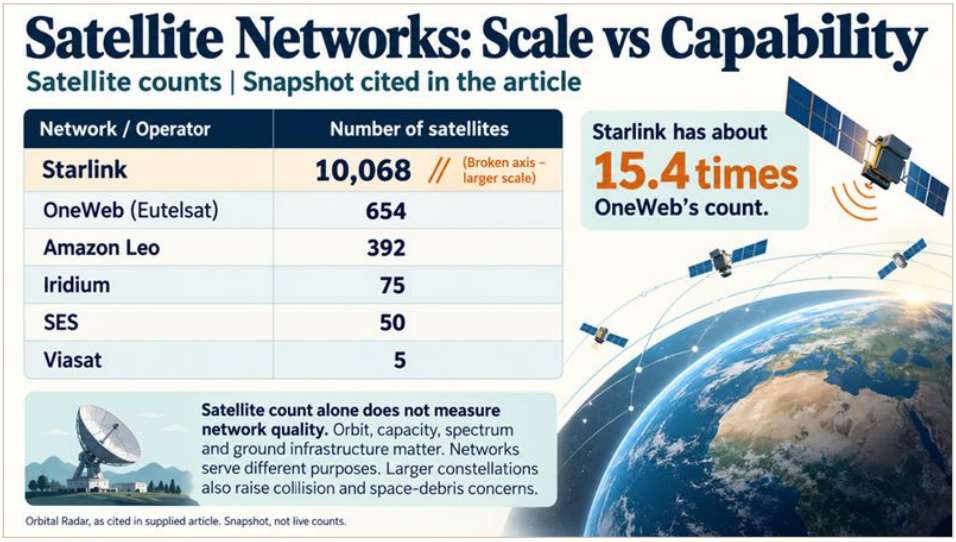
Three distinctions are important:

- **Revenue:** Income from sales before deducting expenses.
- **Operating profit:** Revenue remaining after operating expenses.
- **Capital expenditure:** Investment in assets such as satellites, launch infrastructure and ground equipment.

A profitable business segment can help finance costly research and development. However, **segment profitability does not establish company-wide profitability**, and operating profit is not the same as cash available for investment.

Capacity Expansion and Global Competition

Satellite operators compete through network capacity, spectrum access, coverage, pricing and long-term contracts.



Satellite counts alone cannot establish network superiority. Networks differ in altitude, design, capacity and services offered.

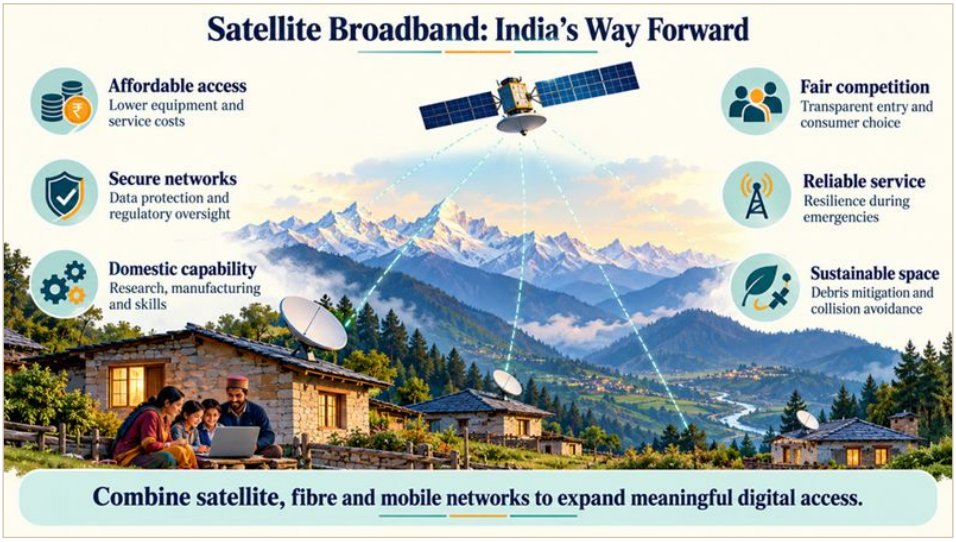
Additional satellites also generate revenue only when providers secure sufficient demand in the regions they serve. Expansion therefore depends on launch capability, ground infrastructure, national authorisations and commercially viable customers.

Large constellations introduce concerns about **orbital congestion, collision risks, space debris and interference with astronomical observations**.

Regulation, Security and Strategic Autonomy

Satellite broadband involves both telecommunications and space infrastructure.

IN-SPACe—the Indian National Space Promotion and Authorisation Centre—authorises relevant space activities. The article reports that Starlink received a telecom licence in June 2025 and IN-SPACe authorisation in July 2025, while describing further clearance requirements for rollout.



The broader policy concerns are:

- **Service continuity:** Maintaining communication during emergencies and geopolitical disputes.
- **Data protection:** Safeguarding information transmitted through networks.
- **Regulatory compliance:** Ensuring effective domestic oversight.
- **Strategic dependence:** Managing reliance on foreign-controlled infrastructure.

India's approach should combine **transparent approvals, fair competition, affordable access and stronger domestic technological capabilities**. Satellite broadband can complement fibre and mobile networks, particularly where conventional infrastructure is difficult to provide.

Mains question: “Satellite broadband is both an economic opportunity and a strategic infrastructure challenge.” Examine. **(250 words)**